

- 11 (a) Using your calculator, find $16\,807^{\frac{1}{5}}$.
 (b) Hence, find the value of x and y in the expression $x^y = 16\,807$.
 (c) Using your calculator, find $64^{\frac{4}{3}}$.
 (d) Hence, find the value of d and e in the expression $d^e = 64$.

Open-ended

- 12 Using your calculator, find at least three ways of writing 343 as a power. When the base number is not exact, use three decimal places. For example, $343 = (18.520)^2$.
 13 Write the expression $n^{\frac{3}{4}} m^{\frac{3}{5}}$ in at least three different ways.

Outside the Square

Problem solving

Lagrange's four-square theorem

Diophantus was a Greek mathematician who lived in the 3rd century. He wrote a famous book called *Arithmetica*. In this book he wrote the following conjecture (a statement that has not been proven).

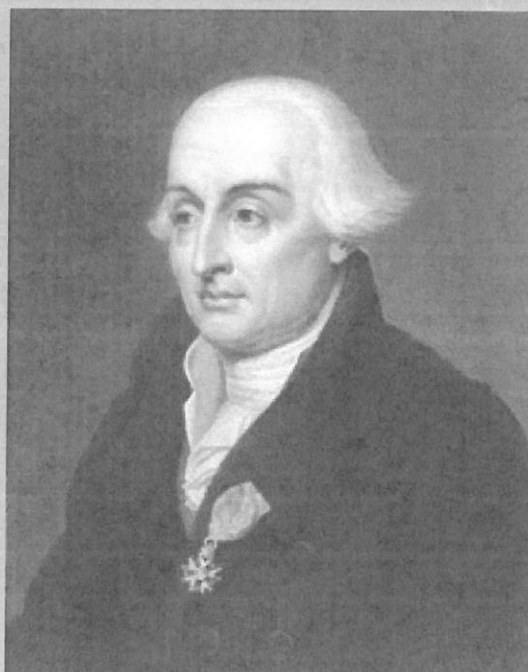
Every whole number is either a square itself or the sum of two, three or four squares.

This conjecture was subsequently proven in 1770 by Joseph-Louis Lagrange.

- (a) Write a square number in each box so that each column and row adds to the total at the end.

			1^2	39
		1^2		27
	1^2			34
1^2				62
36	30	66	54	

- (b) Can any of the totals in the grid be written using fewer square numbers?
 (c) Starting at 1, write each whole number as a sum of four or fewer square numbers. How far can you get in 3 minutes?



Strategy options

- Make a table.
- Break problem into manageable parts.